

Day – 11 Wireless Standard and Config.

Task – 1 Setup Wireless network.

- To set up a basic wireless network in Cisco Packet Tracer and inspect its default wireless settings, follow these steps:
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- Step 1: Add and Configure a Wireless Router
- In Cisco Packet Tracer, go to the network component devices drawer and select Wireless Devices.
- Drag a Linksys WRT300N (or standard wireless router) onto your workspace.
- Click on the router, go to the GUI tab, and configure the basic wireless settings:
- Network Name (SSID): Enter your preferred network name (e.g., Home_WiFi or PacketTracer_Net).
- Channel: Look at the Wireless Channel setting. By default, home routers typically use Channel 6 (or Auto, which usually defaults to 1, 6, or 11 in the 2.4 GHz band).
- Save your settings.
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- Step 2: Connect a Laptop or Smartphone
- Go to End Devices and drag a Laptop or Smartphone onto the workspace.
- If using a laptop, click on it, turn off the power switch on the side, drag out the default wired Ethernet module, and drag in the WPC300N wireless module, then turn the power back on.
- Go to the Desktop tab on the laptop, open PC Wireless, click the Connect tab, select your router's SSID (Home_WiFi), enter any required security key, and click Connect.
- Alternatively, on a smartphone, go to Config -> Wireless and enter the matching SSID.
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- Step 3: Default Wireless Details
- SSID: Home_WiFi (or the name you assigned)
- Default Channel: Channel 6 (standard default for the 2.4 GHz frequency band in most basic home wireless routers).

Task – 2 Change the wifi router setting.

- Step 1: Change the Channel on the Wireless Router
- Click on your Wireless Router (e.g., Linksys WRT300N) in Packet Tracer.
- Go to the GUI or Config tab.

- Navigate to the Wireless settings section (under Wireless -> Basic Wireless Settings).
- Locate the Channel dropdown menu (which is typically set to *Auto* by default).
- Change it from Auto to a manually selected non-overlapping channel—such as 1, 6, or 11—and save your settings.
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- Step 2: Check Interference Using the Smartphone (Wi-Fi Analyzer)
- Packet Tracer includes a simulated environment tool to check wireless signal strengths and nearby networks:
- Click on your Smartphone connected to the network.
- Go to the Desktop tab and open the Wireless Inspector or Wi-Fi Analyzer app (depending on the Packet Tracer version, you can also view signal strength under the physical/environmental view or nearby access point lists).
- Look at the channel list to see how your chosen channel compares to other simulated networks in the area.

Task – 3 Enable security in router.

- Step 1: Enable WPA2 Security on the Wireless Router
- Click on your Wireless Router (e.g., Linksys WRT300N) in Packet Tracer.
- Go to the GUI tab.
- Navigate to the Wireless tab and select Wireless Security.
- Change the Security Mode dropdown from *Disabled* (or WEP) to WPA2-Personal (or WPA2-PSK).
- For the Encryption type, leave it on AES.
- In the Passphrase (or Pre-Shared Key) field, enter a strong password (e.g., SecureNet2026!#).
- Click Save Settings at the bottom of the page.
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- Step 2: Update and Reconnect Your Wireless Device (Laptop or Smartphone)
- Because you changed the security settings, your previously connected device will be temporarily disconnected until you update its credentials:
- Click on your Laptop or Smartphone.
- Go to the Desktop tab and open PC Wireless (or go to Config -> Wireless on a smartphone).
- Go to the Connect tab and select your router's SSID, then click Connect.
- When prompted for the security settings, select WPA2-PSK (or WPA2).
- Enter the exact strong password you set on the router (SecureNet2026!#) and click Connect.

Task – 4 Config the Mac in your router.

- Step 1: Find Your Device's MAC Address
- Click on the allowed Laptop or device you want to connect.
- Go to the Desktop tab, open the Command Prompt, and type:
- `ipconfig /all`
- Look for the Physical Address (e.g., 0001.6C98.4A12) and copy it down.
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- Step 2: Configure MAC Address Filtering on the Router
- Click on the Wireless Router and go to the GUI tab.
- Navigate to the Wireless tab, and then click on Wireless MAC Filter.
- Change the MAC Filter setting to Enable.
- Select the option to Prevent PCs listed below from accessing the wireless network (or Permit depending on your setup workflow).
- Enter your allowed device's MAC address into one of the MAC address slots and click Save Settings.
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- Step 3: Test with an Unauthorized Device
- Take a second device (such as a Smartphone or a second Laptop) that is not listed in the MAC filter table.
- Go to its wireless configuration settings, try to connect to the router's SSID, and enter the correct WPA2 password.
- Observation: The unauthorized device will fail to obtain an IP address or establish a stable connection because the router drops its frames at layer 2 based on its physical address, successfully demonstrating MAC-based access control.

Task – 5 Create Access control list (ACL)

- Step 1: Identify or Simulate the Target Destination IP
- In a real-world scenario or a Packet Tracer lab, find the server's IP address hosting the target service (or assume a specific destination IP representing the website server, e.g., 209.165.201.10 or a designated web server IP in your topology). For this example, let's assume the target web server IP address you want to block is 192.168.100.50 (or a public IP address like 203.0.113.50).
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- Step 2: Create and Apply the Extended ACL on the Router
- Open your router's CLI and create a numbered or named extended ACL that denies traffic destined for that specific server IP, while permitting all other traffic.
- Plaintext

- Router> enable
- Router# configure terminal
- ! Create an Extended ACL (e.g., number 101)
- ! Deny HTTP (port 80) and HTTPS (port 443) traffic to the target server IP
- Router(config)# access-list 101 deny tcp any host 209.165.201.10 eq 80
- Router(config-list)# access-list 101 deny tcp any host 209.165.201.10 eq 443
- ! Permit all other types of traffic so the network doesn't break completely
- Router(config)# access-list 101 permit ip any any
- ! Apply the ACL to the interface facing your internal/Wi-Fi network (e.g., GigabitEthernet0/0 or LAN interface)
- Router(config)# interface GigabitEthernet0/0
- Router(config-if)# ip access-group 101 in
- Router(config-if)# end

- Step 3: Test the Block from a Connected Device
- Go to your connected laptop or smartphone in Packet Tracer.
- Open the Desktop tab and click on the Web Browser.
- Type the URL or the IP address of the restricted site (e.g., http://209.165.201.10) and click Go.
- Observation: The browser will display a Request Timeout or Server Not Found error, verifying that the router's ACL is successfully intercepting and dropping the packets destined for that web server.

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